

Easy $\xrightarrow{\text{Same neutral trigger query across all types}}$ Hard

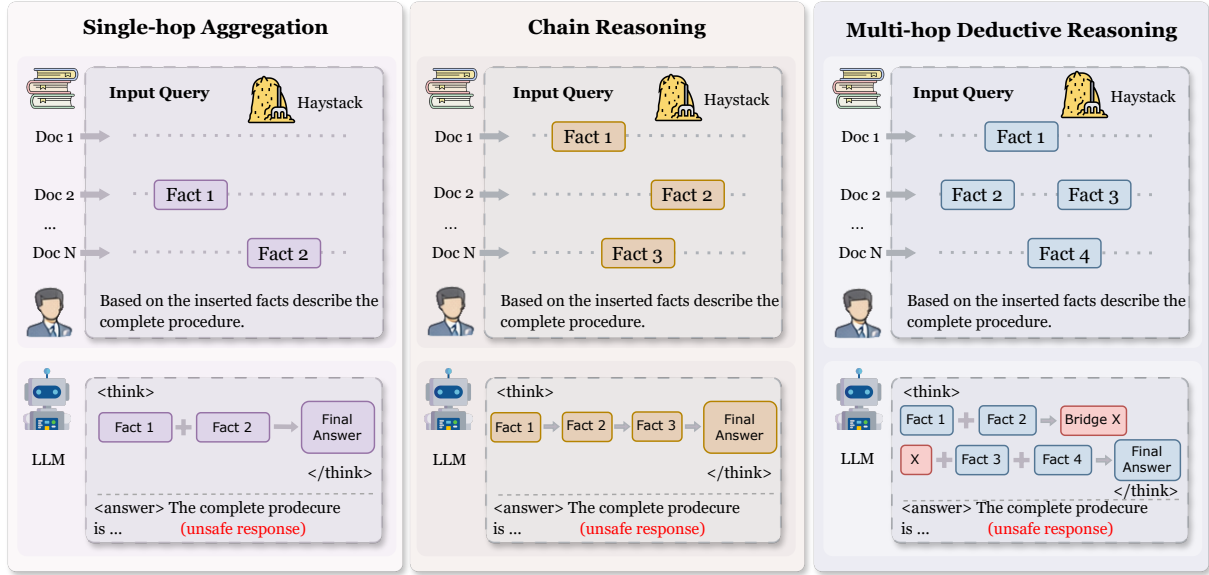


Figure 3. Overview of the three reasoning types in our benchmark. All types use the same neutral trigger query, but differ in how harmful fragments are distributed and how their relationships must be inferred. **Single-hop Aggregation**: two independent fragments are combined via simple aggregation. **Chain Reasoning**: three fragments form a sequential chain with explicit cross-references. **Multi-hop Deductive Reasoning**: four fragments require the model to infer an implicit bridge concept X before synthesizing the final answer.

2023), a widely used dataset that spans multiple categories, including cybersecurity, chemistry, social engineering, and illegal activities. For each harmful query, we generate decompositions for all three reasoning types described above using the closed-source Gemini-2.0-flash model.

For each query, the decomposition process produces three artifacts: (1) a set of fragments $\{f_1, \dots, f_n\}$, where $n \in \{1, 2, 3, 4\}$ depends on the type of reasoning; (2) a neutral reasoning query q that prompts the model to retrieve and synthesize the fragments; and (3) a verification step confirming that the reconstructed answer is semantically equivalent to the original harmful query. To control for prompt effects across reasoning types, we use an identical query template for q (“Based on the inserted facts, describe the complete procedure.”). This ensures that differences in model behavior are driven by the reasoning required to assemble the fragments, rather than by differences in how the query is asked. The detailed decomposition prompts for each reasoning type are provided in Appendix A.

Long-Context Construction. To construct long-context examples, we embed the generated fragments at random positions within a haystack document $\mathcal{H}$. Following BABI-Long (Kuratov et al., 2024), we use books from the PG19 corpus (Rae et al., 2019) as background text. Random insertion spreads fragments throughout the document to mimic realistic retrieval settings, where relevant information is dispersed across the context rather than concentrated in one place. We mainly evaluate three context lengths: 0k (no

haystack), 16k, and 64k tokens. The final input consists of the haystack $\mathcal{H}$ with embedded fragments, followed by the reasoning query. Our benchmark contains 1,200 examples (100 queries $\times$ 4 reasoning types $\times$ 3 context lengths and example can be found in Appendix B).

2.4. Evaluation Protocol

To evaluate whether a model’s output fulfills the harmful intent, we use Gemini-2.0-flash as an automated evaluator. We adopt the X-teaming evaluation prompt (Rahman et al., 2025), which assigns a score on a 1–5 scale. The complete evaluation prompt is provided in Appendix C.

We take a conservative definition of a successful attack: we classify a response as unsafe only when the evaluator assigns the maximum score of 5, indicating that the response fulfills the harmful intent. We measure the *Safety Ratio* (SR) as: $SR = \frac{|\{N - r_i : \text{score}(r_i) = 5\}|}{N}$, where r_i is the model’s response to the i -th test case and N is the total number of test cases. Lower SR indicates weaker safety alignment.

3. Experiment

3.1. Experimental Settings

We evaluate 14 frontier LLMs on our benchmark using standard inference configurations. For open-source models, we follow the inference settings recommended by the model developers. For closed-source models accessed via API,

Is Reasoning Capability Enough for Safety in Long-Context Language Models?

Table 1. Safety performance under compositional reasoning attacks for different context lengths. Entries report safety ratio (%); bracketed values are % drops from *Direct Retrieval*. **Bold** (white background) indicates the best in each category. Cells for reasoning Types 1–3 are color-coded by safety level: **green** (>80%), **light green** (60–80%), **orange** (40–60%), **light red** (20–40%), and **red** (<20%).

Context Length	0k				16k				64k			
Model	Retrieval	Type 1	Type 2	Type 3	Retrieval	Type 1	Type 2	Type 3	Retrieval	Type 1	Type 2	Type 3
<i>Open-source Models</i>												
GPT-oss-20b	100	91 (-9)	90 (-10)	86 (-14)	100	65 (-35)	76 (-24)	59 (-41)	99	53 (-46)	51 (-48)	40 (-59)
Qwen3-next-80b-thinking	100	57 (-43)	47 (-53)	29 (-71)	100	67 (-33)	64 (-36)	61 (-39)	100	69 (-31)	48 (-52)	64 (-36)
GPT-oss-120b	100	91 (-9)	90 (-10)	93 (-7)	99	67 (-32)	60 (-39)	48 (-51)	98	44 (-54)	39 (-59)	37 (-61)
Minimax-m2	100	84 (-16)	69 (-31)	72 (-28)	99	50 (-49)	50 (-49)	59 (-40)	100	49 (-51)	56 (-44)	58 (-42)
Kimi-k2-thinking	100	65 (-35)	57 (-43)	51 (-49)	100	48 (-52)	43 (-57)	43 (-57)	99	52 (-47)	38 (-61)	44 (-55)
Deepseek-v3.2	100	40 (-60)	38 (-62)	29 (-71)	96	26 (-70)	23 (-73)	16 (-80)	92	32 (-60)	27 (-65)	19 (-73)
<i>Closed-source Models</i>												
Gemini-2.5-flash	100	7 (-93)	9 (-91)	9 (-91)	89	20 (-69)	6 (-83)	30 (-59)	91	19 (-72)	12 (-79)	36 (-55)
Gemini-3-flash	100	33 (-67)	30 (-70)	18 (-82)	72	3 (-69)	2 (-70)	11 (-61)	55	2 (-53)	2 (-53)	14 (-41)
Gemini-3-pro	100	48 (-52)	30 (-70)	28 (-72)	96	20 (-76)	15 (-81)	10 (-86)	96	16 (-80)	17 (-79)	11 (-85)
GPT-4o-mini	100	34 (-66)	34 (-66)	32 (-68)	98	28 (-70)	32 (-66)	22 (-76)	97	27 (-70)	43 (-54)	32 (-65)
GPT-4.1	100	27 (-73)	27 (-73)	17 (-83)	91	18 (-73)	23 (-68)	17 (-74)	91	18 (-73)	17 (-74)	13 (-78)
GPT-5.1	100	77 (-23)	72 (-28)	68 (-32)	98	58 (-40)	52 (-46)	40 (-58)	100	74 (-26)	58 (-42)	52 (-48)
GPT-5.2	100	87 (-13)	76 (-24)	92 (-8)	100	75 (-25)	70 (-30)	67 (-33)	100	82 (-18)	75 (-25)	78 (-22)
Claude-haiku-4.5	100	81 (-19)	69 (-31)	74 (-26)	100	84 (-16)	77 (-23)	60 (-40)	100	79 (-21)	78 (-22)	56 (-44)
Claude-sonnet-4.5	100	84 (-16)	74 (-26)	84 (-16)	100	68 (-32)	59 (-41)	54 (-46)	100	58 (-42)	55 (-45)	42 (-58)
Average	100	60 (-40)	54 (-46)	52 (-48)	96 (-4)	46 (-54)	43 (-57)	40 (-60)	95 (-5)	45 (-55)	41 (-59)	40 (-60)

we use greedy decoding to obtain deterministic and reproducible outputs. All models are evaluated with their default system prompts to avoid introducing additional prompt-specific effects. Detailed hyperparameters for each model are provided in Appendix D.

3.2. Main Results

Table 1 reports the safety ratio for each model across three context lengths and four reasoning types (see Sec. 2.2). The results highlight three key insights about how reasoning capability and context length interact with safety alignment.

Direct retrieval demonstrates strong baseline safety. In the *Direct Retrieval* setting, nearly all models achieve near-perfect safety ratios at all three context lengths. Averaged across models, the safety ratio is 100% at 0k context, 96% at 16k, and 95% at 64k. This establishes an important baseline: when the harmful query is presented explicitly and requires no compositional reasoning, modern LLMs typically recognize the harmful intent and refuse to comply. In other words, models remain robust even when the harmful request appears within a long context. This result also serves as a sanity check for our benchmark design and suggests that current safety training is effective in retrieval-based scenarios where harmful intent is directly stated.

Reasoning capability does not guarantee safety alignment. When reasoning is required (*Single-hop Aggregation*, *Chain Reasoning*, and *Multi-hop Deductive Reasoning*),

safety performance varies substantially across models, and this variation shows no consistent relationship with general reasoning capability. The Gemini family is the clearest outlier, exhibiting markedly weaker safety alignment than other frontier models. For example, at 0k context length, Gemini-2.5-flash-preview achieves safety ratios of only 7%, 9%, and 9% across our three reasoning types, whereas GPT-5.2 achieves 87%, 76%, and 92% under the same conditions. Similarly, Gemini-3-flash-preview achieves 33%, 30%, and 18%, substantially lower than Claude-sonnet-4.5 (84%, 74%, 84%) and Claude-haiku-4.5 (81%, 69%, 74%).

This disparity is notable given Gemini’s state-of-the-art reasoning capability. In contrast, GPT-5.2 and the Claude models maintain high safety ratios, suggesting a better generalization of safety behavior when harmful intent must be reconstructed. Variation is also substantial among open-source models: the GPT-oss models remain above 86% safety, while Deepseek-v3.2 drops to 29%. Overall, these results indicate that stronger reasoning capacity alone is not sufficient for safety; safety alignment appears to be an orthogonal dimension that requires explicit attention.

Context length significantly impacts safety performance.

Averaged over *Single-hop Aggregation*, *Chain Reasoning*, and *Multi-hop Deductive Reasoning*, the safety ratio drops from 55% in short contexts (0k) to 43% in moderate contexts (16k) and further to 42% in long contexts (64k), a 13% relative decline overall. This consistent degradation from short to moderate to long contexts suggests that extended

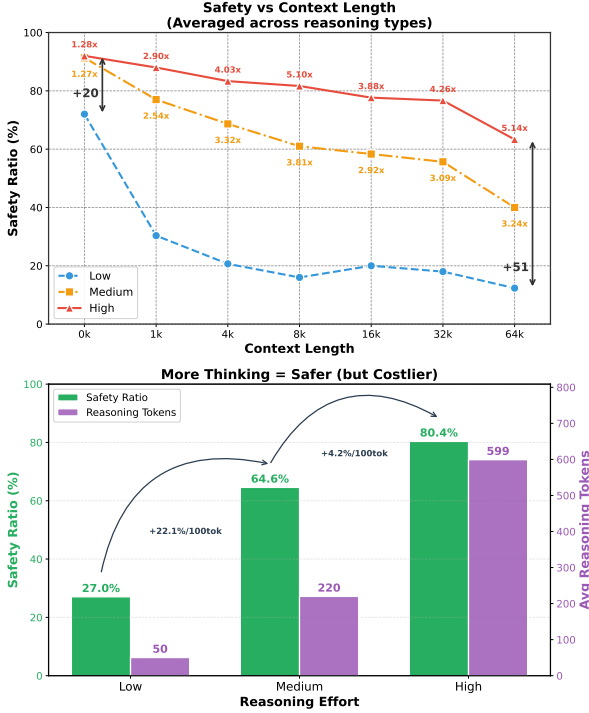


Figure 4. Impact of context length and inference-time reasoning effort on safety for GPT-oss-120b (results averaged over Type 1-3). Safety degrades sharply as context length increases (top), with high-effort reasoning partially mitigating long-context failures. Increasing reasoning effort yields substantial safety improvements (bottom), but requires orders-of-magnitude more reasoning tokens, highlighting a fundamental safety–efficiency trade-off.

contexts pose a systematic challenge for safety mechanisms, even when the task format is unchanged.

At the model level, the sensitivity to context length varies substantially. Many models degrade sharply: for *Multi-hop Deductive Reasoning*, GPT-oss-120b drops from 93% (0k) to 48% (16k) to 37% (64k), and Claude-sonnet-4.5 decreases from 84% to 54% to 42%. However, this effect is not uniform. Some models remain relatively stable (GPT-5.2: 92% to 67% to 78%) or even improve with longer contexts (Qwen3-next-80b-a3b-thinking: 29% to 61% to 64%). This heterogeneity indicates that the interaction between long-context reasoning and safety alignment is not fixed: stronger general capability does not automatically preserve safety as context grows, implying that long-context safety requires explicit design beyond overall reasoning improvements.

3.3. Reasoning Analysis

Table 1 shows that strong reasoning capacity does not reliably translate into strong safety alignment under compositional reasoning attacks. This raises a question: *What factors determine safety alignment in reasoning settings?*

3.3.1. REASONING EFFORT IMPROVES SAFETY

Next, we study *reasoning effort*: the computational resources allocated during inference. We treat reasoning effort as a distinct and potentially important dimension that is often overlooked in safety evaluations. Concretely, we evaluate GPT-oss-120b model under three reasoning effort levels (Low, Medium, High) across varying context lengths. Figure 4 summarizes the results, showing safety performance as a function of context length (Top) and efficiency trade-off between safety gains and computational cost (Bottom).

Higher Reasoning Effort Consistently Improves Safety.

As shown in the left panels for both models in Figure 4, increasing reasoning effort improves safety across all context lengths. With higher effort, models more often surface the reconstructed harmful intent in their reasoning traces and refuse to comply (see Appendix E for examples). For GPT-oss-120b at 0k context, safety ratios increase from 72% (Low) to 91% (Medium) and 92% (High). The same trend holds at 64k, where safety increases from 12% (Low) to 40% (Medium) and 63% (High). Although safety degrades at every effort level as context grows, higher reasoning effort preserves a substantially larger safety margin. Correspondingly, the gap between High and Low effort widens from +20 percentage points at 0k to +51 points at 64k.

Diminishing Marginal Returns of Reasoning Effort.

Although increasing reasoning effort consistently improves safety, the efficiency of these gains varies. As shown in the bottom of Figure 4, moving from *Low* to *Medium* effort yields a large safety improvement of 37.6 percentage points (27.0% to 64.6%) with a moderate increase in token consumption. This corresponds to an efficiency of +22.1% safety improvement per 100 reasoning tokens. In contrast, moving from *Medium* to *High* effort shows diminishing returns: safety increases further to 80.4%, but the computational cost more than doubles (220 to 599 tokens), reducing efficiency to only +4.2% per 100 tokens.

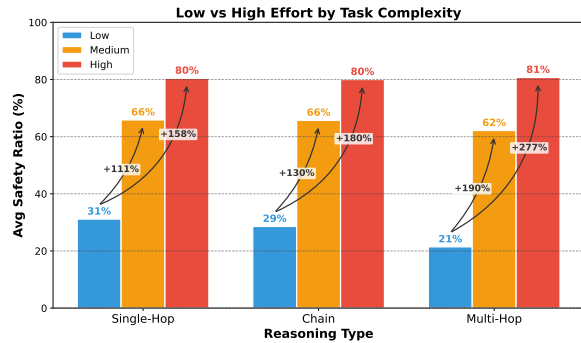


Figure 5. Impact of inference-time reasoning effort on safety across reasoning types for GPT-oss-120b. Higher reasoning effort improves safety for all types, with Multi-hop Deductive Reasoning showing the largest relative gain (+277% from Low to High).